Supplementary Material for "Survival-Time Risk Beyond Low-Order Channel Models: Identifiability and Scheduling"

Ruifeng Zheng, [co-authors]

S1. Overview

This supplement collects material that supports the main paper but does not fit its six pages. Results of the main paper are cited by their numbers there, e.g., Lemma 1, Theorem 1 or Fig. 4; numbering with the prefix "S" refers to this document. Notation follows the main paper. Section S2 illustrates the system model, gives the design of π_L in full and represents the risk of any policy through sampled loss patterns. Section S3 gives complete proofs and additional theory. Sections S4 and S5 report further numerical results behind the claims of the main paper, and Section S6 maps each claim to the script that produces it. Section S5.2 also evaluates a redesign of π_L with the true recovery curve, which the main paper does not include. All result tables and data figures in this supplement are generated from the stored results by `code/make_figs.py` and `code/make_supplement.py`.

S2. System Model and Low-Order Design

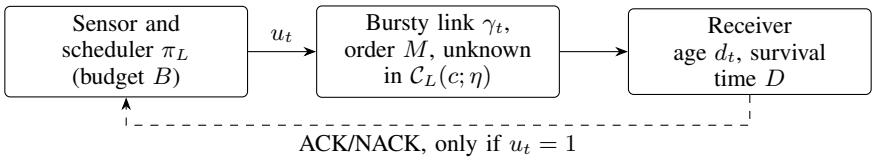


Fig. S1. System model. The latent loss process evolves in every slot; the scheduler observes it only in slots where it transmits.

Fig. S1 shows the setting of Section II. The scheduler π_L minimizes the risk computed under the fitted L-th order model. For $L = 1$ let $q_o = c(o1)/c(o)$ and $f(x) = q_0 + (q_1 - q_0)x$. If the last observed outcome o was observed δ slots ago, the fitted model predicts success with probability $f^{\delta-1}(q_o)$; before any observation it predicts $c(1)$, and the counter δ is not advanced. With $V_{T+1} \equiv 0$, $F(d', z) = 1$ if $d' \geq D$ and $F(d', z) = z$ otherwise, and $q = f^{\delta-1}(q_o)$, the design solves

$$\begin{aligned} V_t(d, r, o, \delta) &= \min\{V^0, V^1\}, \qquad V^0 = F\big(d{+}1, V_{t+1}(d{+}1, r, o, \delta{+}1)\big), \\ V^1 &= q\, V_{t+1}(0, r{-}1, 1, 1) + (1{-}q)\, F\big(d{+}1, V_{t+1}(d{+}1, r{-}1, 0, 1)\big), \end{aligned} \tag{S1}$$

where V^1 requires $r \geq 1$ and ties are broken toward idling. The nominal risk R_{nom} is $V_1(0, B, \varnothing, 0)$. If $B \geq T$, attempting in every slot minimizes the age pathwise on every channel, since an idle slot can only delay a reset. Where a failure is still possible, idling leads, with positive probability under the fit, to a failure that an attempt would avoid, because the fit is η-positive whenever $\mathcal{C}_L(c;\eta)$ is nonempty. Hence π_L attempts in exactly these slots, and its risk equals that of dense attempts on every channel. The same recursion with an arbitrary recovery curve $\Pr(\gamma_{t+\delta} = 1 \mid o)$ in place of $f^{\delta-1}(q_o)$ defines the redesign π_{rec} of Section S5.2.

Proposition S1 (Sampled-pattern representation)**.** *For every deterministic policy π with budget B under ACK/NACK feedback, there is a finite set $\mathcal{H}_\pi$ of attempt-slot tuples $\mathbf{s} = (s_1 < \cdots < s_j)$ and outcome tuples $\mathbf{o} \in \{0,1\}^j$, determined by π alone, such that for every channel $\mathcal{M}$*

$$R_{\mathcal{M}}(\pi) = \sum_{(\mathbf{s},\mathbf{o}) \in \mathcal{H}_\pi} \Pr_{\mathcal{M}}(\gamma_{s_1} = o_1, \ldots, \gamma_{s_j} = o_j). \tag{S2}$$

Proof: The controller's information, the age and the remaining budget are functions of the attempted outcomes only. Enumerating the ACK/NACK histories up to the failure slot partitions $\{\tau \leq T\}$ into disjoint cylinder events on attempted slots. ∎

The risk is therefore a linear functional of gapped pattern probabilities, and the task selects which patterns enter. It is identifiable from c if all of them are. This sufficient condition fails in general: with dense attempts and $D > L + 1$, for instance, every failure pattern fixes at least D consecutive outcomes. The closed-loop decomposition Eq. (15) makes the selected patterns explicit.

S3. Proofs and Additional Theory

S3.1 Proof of Lemma 1

A stationary M-th order chain with positive context law is equivalent to a shift-consistent $(M{+}1)$-block law ν, i.e., one whose left and right M-marginals coincide (both equal the context law μ): given such ν, the transition probabilities $p(w) = \nu(w1)/\mu(w)$ started from μ define a stationary chain with block law ν, and conversely. For $M = L{+}1$ the $(L{+}1)$-block law of the chain is μ itself, so compatibility means that both M-marginals of ν equal c. For a middle word $m \in \{0,1\}^L$ the four entries $\nu(a\,m\,a')$, $a, a' \in \{0,1\}$, form a 2×2 table with row sums $c(am)$ and column sums $c(ma')$; distinct middle words involve disjoint entries. Each table has one free entry $x_m = \nu(0m0)$, and the others follow as stated in the lemma. Nonnegativity yields the Fréchet interval $\max\{0, c(m0) - c(1m)\} \le x_m \le \min\{c(0m), c(m0)\}$. The constraint $\eta \le p(am) \le 1 - \eta$ reads $x_m \in [\eta c(0m), (1-\eta)c(0m)]$ for $a = 0$ and $c(1m) - c(m0) + x_m \in [\eta c(1m), (1-\eta)c(1m)]$ for $a = 1$; the intersection is $[x_m^{\mathrm{lo}}, x_m^{\mathrm{hi}}]$ and implies nonnegativity. Since the tables are independent, the class is the product of these intervals, and the map is affine.

The entropy rate of the chain is $\bar{H} = H(\gamma_{M+1} \mid \gamma_{1:M}) = H(\gamma_{L+2} \mid \gamma_1, \gamma_{2:L+1}) \le H(\gamma_{L+2} \mid \gamma_{2:L+1})$, and the right side depends only on c. Equality holds iff γ_1 and γ_{L+2} are conditionally independent given every middle word, i.e., $\nu(a\,m\,a') = c(am)c(ma')/c(m)$, which gives x_m^{nom}; this is the fitted L-th order model, whose transition $\Pr(a' \mid m) = c(ma')/c(m)$ ignores a. Uniqueness follows from strict concavity of the conditional entropy in each table. In the running example, $x_0 = \nu(000) \in [\frac{11}{250}, \frac{57}{500}]$ and $x_1 = \nu(010) \in [\frac{1}{250}, \frac{11}{250}]$, and the class is parameterized by (ρ, b) through $p(00) = 1 - \rho$, $p(01) = b$, $p(10) = 1 - \frac{3}{2}(1-\rho)$, and $p(11) = 1 - \frac{b}{9}$.

S3.2 Proof of Theorem 1 and Eq. (12)

Continue the age process past D (the schedule is open-loop, so attempts do not depend on failure) and let $I_t = \mathbf{1}\{d_t = D\}$. The age increases by one per slot until it is reset, so it cannot jump over D, and $\{\tau \le T\} = \{\sum_{t \le T} I_t \ge 1\}$. Moreover, $I_t = 1$ iff the last reset before t occurred at $t - D \in \mathcal{S}_0$ and no attempt in $(t-D, t]$ succeeded, i.e., iff A_{t-D} occurs; hence $\mathbb{E}\sum_t I_t = \Lambda_{\mathcal{S}}$, and the union bound gives $R \le \Lambda_{\mathcal{S}}$. This step uses neither the Markov property nor stationarity.

For two hits $t < t'$, the reset at $t' - D$ comes after t: if $t' - D \le t$, the age at t would be $t - (t' - D) < D$ or 0. Hence $t' - D > t$, so I_t is measurable with respect to $\mathcal{F}_{t'-D-1} = \sigma(S_1, \gamma_{1:t'-D-1})$, and $t' - D \in \mathcal{S}$ because $t' - D > t \ge D \ge 1$. By the Markov property, $\mathbb{E}[I_{t'} \mid \mathcal{F}_{t'-D-1}] = \Pr(A_{t'-D} \mid S_{t'-D}) \le \kappa_{\mathcal{S}}$. For fixed $t = s + D$ there are at most $J_{\mathcal{S}}$ admissible t', so $\sum_{t<t'} \mathbb{E}[I_t I_{t'}] \le J_{\mathcal{S}} \kappa_{\mathcal{S}} \Lambda_{\mathcal{S}}$, and the second Bonferroni inequality $\Pr(\cup E_i) \ge \sum \Pr(E_i) - \sum_{i<j} \Pr(E_i E_j)$ yields the lower bound.

For dense attempts, $A_0 = \{0^D\}$ and $A_s = \{\gamma_s = 1\} \cap \{0^D \text{ on } (s, s+D]\}$ for $s = 1, \dots, n$. Since $\Pr(0^{M+1}) = x_{0^L} = \rho\, c(0^M)$ and every further loss after M losses has probability ρ, $\Pr(0^D) = c(0^M)\rho^k$ and, by stationarity, $\Pr(10^D) = \Pr(0^D) - \Pr(0^{D+1}) = c(0^M)(1-\rho)\rho^k$. Given $S_s = w$, A_s requires a success followed by D losses, so $\Pr(A_s \mid w) \le \rho^k \max_{v: v_M = 1} \Pr(0^M \mid v)$, and $J = (T - 2D)^+ \le n$; this gives $J\kappa \le \theta_{\mathcal{M}} \Lambda$. In the running example ($M = 2$), the contexts ending in a success are 01 and 11, and $\Pr(00 \mid v) = (1 - p(v))(1 - p(10))$ with $1 - p(10) = \frac{3}{2}(1-\rho)$, $1 - p(01) = 1 - b$ and $1 - p(11) = \frac{b}{9}$; dividing by $c(00)(1-\rho) = 0.12(1-\rho)$ gives $\theta_{\mathcal{M}} = 12.5 \max\{1 - b, \frac{b}{9}\} \le 6.9$.

S3.3 Proof of Theorem 2

Throughout, $k \ge 1$, $n = T - D \ge 0$ and $0 < \rho_{\mathrm{lo}} \le \rho_{\mathrm{hi}} < 1$, because every compatible channel has $\eta \le \rho \le 1 - \eta$. With $B \ge T$, the risks $R^{\pm}$ and R_{nom} are those of dense attempts.

Monotonicity and unimodality. With $\varphi(\rho) = \rho^k[1{+}n(1{-}\rho)]$, $\varphi'(\rho) = \rho^{k-1}[k(n{+}1){-}(k{+}1)n\rho]$. The bracket is nonincreasing in ρ and equals $k - n$ at $\rho = 1$. If $n \le k$, it is positive on $(0,1)$, so φ is increasing and ρ_{hi} maximizes $\Lambda = c(0^M)\varphi(\rho)$. If $n > k$, it vanishes at $\rho^\star = k(n+1)/[(k+1)n] \in (k/(k+1), 1)$, so φ increases on $(0, \rho^\star]$ and decreases on $[\rho^\star, 1)$. On $[\rho_{\mathrm{lo}}, \rho_{\mathrm{hi}}]$, the maximum is therefore attained at the point closest to $\rho^\star$, i.e., at $\rho_w = \min\{\max\{\rho^\star, \rho_{\mathrm{lo}}\}, \rho_{\mathrm{hi}}\}$, and the minimum at an endpoint, since a unimodal function has no interior minimum.

Most persistent channel. By the first two claims, ρ_{hi} maximizes Λ for a given n if and only if $n \le k$ or $\rho^\star(n) \ge \rho_{\mathrm{hi}}$. For $n > k$, $\rho^\star(n) = \frac{k}{k+1}(1 + \frac{1}{n})$ decreases strictly in n toward $k/(k+1)$ without attaining it. Hence $\rho^\star(n) \ge \rho_{\mathrm{hi}}$ for all $n > k$ if and only if $k/(k+1) \ge \rho_{\mathrm{hi}}$, i.e., $k \ge \rho_{\mathrm{hi}}/(1 - \rho_{\mathrm{hi}})$. As $T \ge D$ ranges over all $n \ge 0$, this proves the third claim.

Growth with D. For $\rho' < \rho_{\mathrm{hi}}$, let $x = 1 - \rho_{\mathrm{hi}}$ and $y = 1 - \rho'$, so $0 < x < y$. Then $\varphi(\rho_{\mathrm{hi}})/\varphi(\rho') = (\rho_{\mathrm{hi}}/\rho')^k(1{+}nx)/(1{+}ny)$, and $y(1+nx) - x(1+ny) = y - x > 0$ gives $(1+nx)/(1+ny) > x/y$, which is the stated inequality. For the ratios, let C_{hi} and C_{lo} be compatible channels with $\rho = \rho_{\mathrm{hi}}$ and $\rho = \rho_{\mathrm{lo}}$, and let C_{nom} be the fitted model, which is compatible whenever the class is nonempty. By Theorem 1 and $J\kappa \le \theta_{\mathcal{M}}\Lambda$ from the previous subsection, every compatible channel satisfies $\Lambda(1 - \theta_{\mathcal{M}}\Lambda) \le R \le \Lambda$ with $\Lambda = c(0^M)\varphi(\rho)$. Since $R^+ \ge R(C_{\mathrm{hi}})$ and $R_{\mathrm{nom}} = R(C_{\mathrm{nom}})$,

$$\frac{R^+}{R_{\mathrm{nom}}} \ge (1 - \theta_{\mathrm{hi}}\Lambda_{\mathrm{hi}}) \frac{\varphi(\rho_{\mathrm{hi}})}{\varphi(\rho_{\mathrm{nom}})} \ge (1 - \theta_{\mathrm{hi}}\Lambda_{\mathrm{hi}}) \Big(\frac{\rho_{\mathrm{hi}}}{\rho_{\mathrm{nom}}}\Big)^k \frac{1 - \rho_{\mathrm{hi}}}{1 - \rho_{\mathrm{nom}}},$$

where θ_{hi} and Λ_{hi} refer to C_{hi}. Since $R^- \le R(C_{\mathrm{lo}})$, the same bound holds for R^+/R^- with ρ_{lo} in place of ρ_{nom}. Apart from the factor $1-\theta_{\mathrm{hi}}\Lambda_{\mathrm{hi}}$, the bound does not depend on n, and $k = D - M$. Each additional slot of D therefore multiplies it by $\rho_{\mathrm{hi}}/\rho_{\mathrm{nom}}$ or $\rho_{\mathrm{hi}}/\rho_{\mathrm{lo}}$, as long as $\theta_{\mathrm{hi}}\Lambda_{\mathrm{hi}}$ stays small, which is the rare-failure regime of the theorem.

Fitted model. The fitted model is the worst compatible channel only if $\rho_{\mathrm{nom}} = \rho_w$; in the running example this happens, e.g., for $k = 1$, $n = 5$ ($D = 3$, $T = 8$), where $\rho^\star = \frac{6}{10} = \rho_{\mathrm{nom}}$. Generically the fitted model is not the worst case.

Required precision. $\partial \log \Lambda / \partial \rho = k/\rho - n/[1 + n(1-\rho)]$, which is 14.3 at $\rho_{\mathrm{nom}} = 0.6$ for $T = 60$, $D = 12$; an error of $\Delta\rho$ multiplies the risk by about $e^{14.3\,\Delta\rho}$, so a factor-two accuracy requires $|\Delta\rho| \lesssim 0.049$, and the required precision grows linearly in D. In the running example, $\rho = \Pr(0 \mid 00)$ is estimated from the $\approx c(00)N = 0.12N$ occurrences of the context 00 in N slots; ignoring the dependence between occurrences, a two-standard-error accuracy of 0.049 needs $N \approx 3.4 \times 10^3$ slots. The estimation problem becomes hard only when the order is unknown.

S3.4 Decimated run persistences

For M-th order chains and $g \ge M$, the run persistences $\rho_j^{(g)}$ of Proposition 1 converge to σ_g as $j \to \infty$, because $\Pr^{(g)}(0^j) = \alpha^\top Q_g^{j-1}\mathbf{1}$ for a fixed nonnegative vector α and Q_g is entrywise positive. For C_1 and C_2, the 2-decimated persistences start at $\rho_1^{(2)} = 0.284$ and 0.44 and remain between 0.2326 and 0.284 and between 0.4293 and 0.44, respectively, for $j \le 60$, where they match σ_2 to eight digits. Hence C_1 is below C_2 at every depth.

S3.5 Decay rates of spaced attempts

Proposition S2 (Decay exponents)**.** *With Q_g and σ_g as in Section IV, $\sigma_1 = \rho$, $\sigma_g \ge \rho^g$, and Q_g is entrywise positive for $g \ge M$. If $\log(1 + T - D) = o(D)$, and $g = 1$ or $g \ge M$, then Eq. (14) holds.*

Proof: Let $D_g = \lfloor D/g \rfloor$. For $s \in \mathcal{S}_g$, define $\alpha_s(w) = \Pr(\gamma_s = 1, \gamma_{s+g} = 0, S_{s+g+1} = w)$ on loss-ending contexts; then $\Pr(A_s) = \alpha_s^\top Q_g^{D_g - 1}\mathbf{1}$, and similarly for A_0. By stationarity α_s does not depend on s, and $\Lambda_{\mathcal{S}_g}$ has at most $n/g + 1$ such terms. For $g \ge M$, Q_g is entrywise positive, so Perron–Frobenius gives $Q_g^j = \sigma_g^j(\xi\zeta^\top + o(1))$ with positive ξ, ζ, hence $\Pr(A_s) = \Theta(\sigma_g^{D_g})$ and $\log \Lambda_{\mathcal{S}_g} = D_g \log \sigma_g + O(\log n) + O(1)$. Since every context has η-positive success probability, Q_g is strictly substochastic and $\sigma_g < 1$. The second-order term satisfies $J\kappa \le (n/g)\max_w e_w^\top Q_g^{D_g - 1}\mathbf{1} = O(n\sigma_g^{D_g}) \to 0$ when $\log n = o(D)$, so $\log R = \log \Lambda + o(1)$ and $\frac{1}{D}\log R \to \frac{1}{g}\log \sigma_g$. For $g = 1$, Q_1 is not primitive, but Eq. (12) gives Λ in closed form and $\frac{1}{D}\log \Lambda \to \log \rho$. In the loss graph of Q_1 the only cycle is the self-loop at 0^M, so $\sigma_1 = \rho$. Finally, $\sigma_g \ge Q_g(0^M, 0^M) \ge \rho^g$, since the spectral radius of a nonnegative matrix dominates every diagonal entry, and for $g \ge M$ all $g - 1$ intermediate outcomes are free, so every loss-ending context is reached with η-positive probability. ∎

For $M = 2$, $g = 2$ and contexts ordered $(00, 10)$,

$$Q_2 = \begin{pmatrix} \rho^2 & (1-\rho)(1-b) \\ (1 - p(10))\rho & p(10)(1-b) \end{pmatrix}, \tag{S3}$$

whose second column (loss–success–loss) holds the relapse terms. For C_1 and C_2, $\sigma_2 = 0.238$ and 0.433; exact risks at $n = 48$ give log-ratio slopes 0.1823 and 0.2986 per unit of D, against $\log(\rho(C_1)/\rho(C_2)) = 0.1823$ and $\frac{1}{2}\log(\sigma_2(C_2)/\sigma_2(C_1)) = 0.2986$, and the reversal holds for every even D from 4 to 40. For large D this is a criterion: if $\rho(C_1) > \rho(C_2)$ but $\sigma_g(C_1) < \sigma_g(C_2)$, with $g \ge M$ and $\log n = o(D)$, dense and g-spaced attempts rank the two channels oppositely, with log-ratios growing at these slopes.

S3.6 Closed-loop decomposition

Proposition S3 (Closed-loop decomposition)**.** *For every deterministic policy π under ACK/NACK feedback, $R(\pi) = \Lambda_\pi$ with Λ_π as in Eq. (15).*

Proof: After a success at s the controller state is a function of the observation history h up to s; for π_L it is $(d, o, \delta) = (0, 1, 1)$ together with the remaining budget r. As long as the attempts fail, the observations are NACKs, so the decisions in $(s, s + D]$ are a deterministic function of (s, h), the retry pattern $\mathcal{R}_\pi(s, h)$. If $\tau \le T$, the last success before τ, or the start $s = 0$, is unique, precedes any violation and is followed by losses at all attempts of $\mathcal{R}_\pi(s, h)$, and $\tau = s + D$. Conversely, a success at s with $s + D \le T$ before any violation, followed by such losses, gives $\tau = s + D$. The events $E_{s,h,w} \cap \{\gamma_j = 0 \ \forall j \in \mathcal{R}_\pi(s, h)\}$ therefore partition $\{\tau \le T\}$. Since $E_{s,h,w}$ is determined by $\gamma_{1:s}$ and fixes $S_{s+1} = w$, the Markov property gives $\Pr(E_{s,h,w}, \gamma_{\mathcal{R}_\pi(s,h)} = 0) = \Pr(E_{s,h,w}) \Pr(\gamma_{\mathcal{R}_\pi(s,h)} = 0 \mid S_{s+1} = w)$, and summing over the partition yields $R(\pi) = \Lambda_\pi$. For the start ($s = 0$), h is empty, $r = B$ and $S_1 \sim \mu$. ∎

Unlike $\Lambda_{\mathcal{S}}$ in Theorem 1, Λ_π weights each retry pattern by the probability of a success before any violation, which depends on π and follows from the same forward recursion as R. The identity therefore shows which loss patterns enter the risk rather than giving a closed form. Continuing π after a violation, e.g., by attempting until the next success, would add later successes and turn it into the upper bound $R(\pi) \le \Lambda_\pi$. For π_L ($T = 60$, $B = 15$, $D = 12$) on a 9×5 grid of the class, the computed Λ_π matches R to a relative 10^{-14}.

S3.7 No task-agnostic burstiness index

Corollary S1. *No function ψ on $\mathcal{C}_L(c;\eta)$, such as the burst-length tail, a lag-g correlation with $g > L$, ρ or a Gilbert–Elliott parameter, satisfies $R_{\mathcal{M}}(\pi_L) < R_{\mathcal{M}'}(\pi_L) \Rightarrow \psi(\mathcal{M}) < \psi(\mathcal{M}')$ for every task (T, D, B).*

Proof: Item 1) of Proposition 2 forces $\psi(C_2) < \psi(C_1)$, and item 3) forces $\psi(C_1) < \psi(C_2)$. ∎

S3.8 Order mismatch $M \geq L + 2$

Remark S1. *For $M \geq L + 2$ the compatible class is a general polytope, and $y = \Pr(0^M)$ is no longer fixed by c. With $x = \Pr(0^{M+1})$, Eq. (12) generalizes to $\Lambda = y\,\varphi(x/y)$, which is homogeneous of degree one in (y, x). For fixed $\rho = x/y$, Λ is proportional to y, so the maximum over the class lies where a ray of slope ρ leaves the two-dimensional projection of the polytope onto (y, x), i.e., on its boundary; the projection follows from linear programs. For the running example with $M = 3$ instead of $M = 2$ ($T = 60$, $D = 12$), $y \in [0.044, 0.114]$ (nominal 0.072), and a grid search over the projected polygon gives $\Lambda_{\max} = 0.254$ (0.253 for $M = 2$) at $y = 0.112$ and $x/y = 0.928$, where the exact dense risk is 0.240. Since the $M = 3$ class contains the $M = 2$ class, whose dense worst case is 0.242, the $M = 3$ dense worst case lies in $[0.242, 0.254]$. The closed-loop worst case over the polytope remains open.*

S4. Additional Results for the Running Example

S4.1 Dense attempts

Table S1 compares the worst-case persistence of Theorem 2 with exact maximizers over a grid of 1201 values of ρ at $b = 0.7$. The maximizer of Λ matches ρ_w to grid resolution, and the maximizer of R deviates slightly where R saturates; over all cases, the two-sided bound is violated by at most 1.4×10^{-16}. Fig. S2 shows the phase diagram: for long missions the worst channel is interior, and once $D \geq D^\star = 21$ the cap $\rho_{\text{hi}} = 0.95$ binds for every mission length. The right panel shows that the risk ratios R^+/R_{nom} and R^+/R^- grow geometrically in D, as predicted by Λ, until R^+ saturates. Table S2 repeats the comparison for a third-order class ($L = 2$, $M = 3$), a four-dimensional box with $\rho \in [0.488, 0.950]$ and $\rho_{\text{nom}} = 0.650$. Its column "spread" is the largest relative change of the dense risk when the three relapse coordinates are redrawn at fixed ρ; it is at most 2.7%. Over random channels and dense, periodic and irregular open-loop schedules ($T = 60$, $D = 12$), the two-sided bound of Theorem 1 was never violated, and its gap $(\Lambda_{\mathcal{S}} - R)/R$ never exceeded 16%.

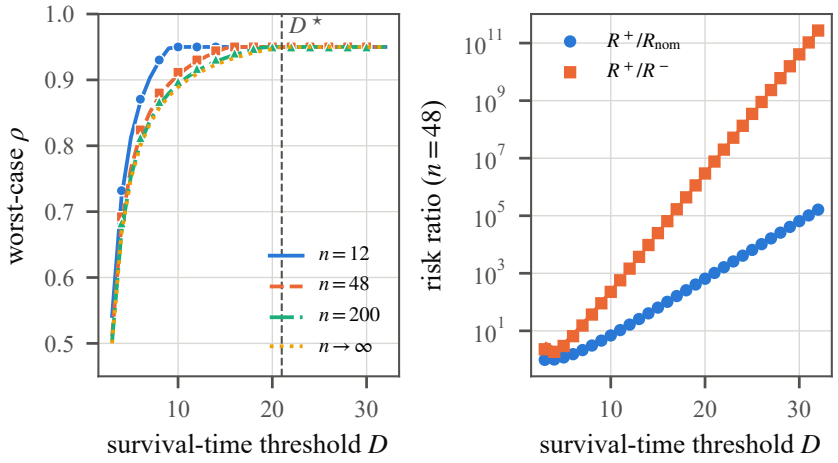


Fig. S2. Left: worst-case persistence ρ_w (lines: Theorem 2; markers: exact risk maxima) versus D for $n = T - D$; the cap $\rho_{\text{hi}} = 0.95$ is active for all n beyond $D^\star = 21$. Right: exact risk ratios for $n = 48$ (markers) and their predictions from Λ (lines).

S4.2 Budgeted attempts, certificates and the positivity bound

Table S3 is the closed-loop sweep over the budget summarized in Fig. 3 ($T = 60$, $D = 12$; a grid of 47×21 channels in (ρ, b) with resolution 0.013 in ρ). As B decreases, the worst channel moves toward less persistent channels with frequent relapses ($b = 0.45$), and the best channel, at ρ_{lo}, moves from $b = 0.45$ ($B = 60$) to $b = 0.95$ (all smaller budgets). At $\rho = \frac{11}{30}$, switching b from 0.95 to 0.45 leaves the dense risk nearly unchanged but raises the risk of periodic attempts every four slots by $3.88\times$, every two slots by $34.6\times$, and that of π_L with $B = 15$ by $9.05\times$; at $\rho = 0.95$ the same switch changes the dense, every-four-slots and π_L risks by at most 19%. Table S4 lists the certificates discussed in Section V ($T = 60$). For dense attempts, its worst cases add a search over 2001 values of ρ at $b \in \{0.45, 0.95\}$ to the grid of Table S3, which raises the value at $D = 12$ from 0.241 to 0.242. The interval relaxation, i.e., the rectangular robust MDP, lets each context's success

TABLE S1
DENSE ATTEMPTS OVER THE COMPATIBLE CLASS OF THE RUNNING EXAMPLE

T	D	ρ_w (Thm.)	argmax Λ	argmax R	R^+	R^-	R_{nom}	R^+/R_{nom}
6	6	0.950	0.950	0.950	9.8×10^{-2}	2.2×10^{-3}	1.6×10^{-2}	6.3
12	6	0.933	0.933	0.933	1.3×10^{-1}	1.0×10^{-2}	5.3×10^{-2}	2.4
30	6	0.833	0.833	0.840	2.7×10^{-1}	3.5×10^{-2}	1.6×10^{-1}	1.7
60	6	0.815	0.815	0.822	4.7×10^{-1}	7.4×10^{-2}	3.1×10^{-1}	1.5
120	6	0.807	0.807	0.815	7.2×10^{-1}	1.5×10^{-1}	5.3×10^{-1}	1.4
30	10	0.933	0.933	0.935	1.6×10^{-1}	5.4×10^{-4}	1.8×10^{-2}	8.8
60	10	0.907	0.907	0.910	2.9×10^{-1}	1.3×10^{-3}	4.2×10^{-2}	6.8
200	10	0.894	0.894	0.898	6.8×10^{-1}	4.7×10^{-3}	1.4×10^{-1}	4.7
100	15	0.939	0.939	0.942	3.0×10^{-1}	1.4×10^{-5}	5.5×10^{-3}	55
300	15	0.932	0.932	0.935	6.6×10^{-1}	4.7×10^{-5}	1.8×10^{-2}	37
100	24	0.950	0.950	0.950	1.8×10^{-1}	1.5×10^{-9}	5.0×10^{-5}	$3.6k$
400	24	0.950	0.950	0.950	5.6×10^{-1}	7.4×10^{-9}	2.4×10^{-4}	$2.4k$

TABLE S2
DENSE ATTEMPTS OVER THE THIRD-ORDER CLASS ($L = 2$, $M = 3$)

T	D	ρ_w (Thm.)	argmax R	R^+	R^-	R_{nom}	spread
8	8	0.950	0.950	4.6×10^{-2}	1.7×10^{-3}	7.0×10^{-3}	0.0%
60	8	0.849	0.847	2.2×10^{-1}	4.5×10^{-2}	1.3×10^{-1}	2.7%
200	8	0.838	0.842	5.7×10^{-1}	1.5×10^{-1}	3.9×10^{-1}	2.4%
100	12	0.910	0.913	2.1×10^{-1}	4.4×10^{-3}	3.9×10^{-2}	1.6%
400	12	0.902	0.899	6.2×10^{-1}	1.9×10^{-2}	1.6×10^{-1}	1.5%
150	30	0.950	0.950	1.0×10^{-1}	1.5×10^{-8}	2.3×10^{-5}	0.5%

probability range over the projection of the box and be chosen adversarially in every slot, and its adversary also chooses the initial context; the time-invariant corner fixes the worst corner of these intervals for the whole mission and starts in its stationary law. Parameter lifting [1] also relaxes the coupling of the box, but its bounds tighten as the parameter space is subdivided. Table S5 varies the positivity bound η ($T = 60$, $D = 12$, $R_{\mathrm{nom}} = 0.0146$) and maximizes over ρ at the midpoint of the range of b, which gives 0.240 instead of 0.242 at $\eta = 0.05$. For $\eta \le 0.05$, the dense worst case is interior and barely moves. For $\eta = 0.1$, $\rho_{\mathrm{hi}} = 0.9$ lies below $\rho^\star = 0.928$, so the worst case moves to ρ_{hi}, as Theorem 2 predicts.

TABLE S3
CLOSED-LOOP RISK OF π_L OVER THE COMPATIBLE CLASS

B	R_{nom}	R^+	at (ρ, b)	R^-	R^+/R_{nom}
60	1.5×10^{-2}	0.241	(0.925, 0.95)	1.7×10^{-4}	16.5
30	2.2×10^{-2}	0.254	(0.925, 0.45)	4.7×10^{-4}	11.4
20	3.7×10^{-2}	0.269	(0.912, 0.45)	1.7×10^{-3}	7.3
15	6.0×10^{-2}	0.286	(0.899, 0.45)	6.7×10^{-3}	4.8
12	9.3×10^{-2}	0.303	(0.887, 0.45)	2.2×10^{-2}	3.3

TABLE S4
RISK CERTIFICATES FOR THE RUNNING EXAMPLE

Task	R_{nom}	R^+ (grid)	box $\Lambda_{\max}$	interval relax.	time-inv. corner
dense, $D = 12$	1.5×10^{-2}	0.242	0.253	0.973	0.967
dense, $D = 20$	2.1×10^{-4}	0.142	0.143	0.795	0.782
$B = 15$, $D = 12$	6.0×10^{-2}	0.286	–	0.985	0.982

S4.3 Exact certificate of Proposition 2

The risks of Proposition 2 were computed in exact rational arithmetic (`fractions.Fraction`), including the design of π_L. The full fractions are:

TABLE S5
SENSITIVITY OF THE DENSE WORST CASE TO η

η	$[\rho_{\mathrm{lo}}, \rho_{\mathrm{hi}}]$	worst ρ	R^+	R^-	R^+/R_{nom}
0.01	$[0.340, 0.990]$	0.931	0.239	8.1×10^{-5}	16.3
0.02	$[0.347, 0.980]$	0.931	0.239	9.7×10^{-5}	16.4
0.05	$[0.367, 0.950]$	0.931	0.240	1.7×10^{-4}	16.5
0.1	$[0.400, 0.900]$	0.900	0.230	3.7×10^{-4}	15.8

- dense attempts, $R(C_1) \approx 9.096922\times10^{-4}$:
  `28764046942402247011864795479079374618670119590699773794325455937275452419409017556257663661031306070096308800072783045878393774943/31619537076459609040430879785135516613081563264131546020507812500000000000

TABLE S7
GILBERT–ELLIOTT CHANNELS SHARING THE GILBERT FIT OF THE RUNNING EXAMPLE

(e_G, e_B)	memory	ρ	b	ρ_∞	dense	proj.	π_L, B=15	proj.	π_{rec} / genie
$(0, 1)$	0.50	0.600	0.900	0.600	1.46×10^{-2}	1.46×10^{-2}	5.97×10^{-2}	5.97×10^{-2}	6.0×10^{-2} / 2.6×10^{-2}
$(0, 0.8)$	0.67	0.600	0.733	0.600	1.46×10^{-2}	1.46×10^{-2}	8.25×10^{-2}	8.20×10^{-2}	7.6×10^{-2} / 5.9×10^{-2}
$(0, 0.7)$	0.80	0.600	0.600	0.600	1.46×10^{-2}	1.46×10^{-2}	1.06×10^{-1}	1.02×10^{-1}	8.4×10^{-2} / 7.7×10^{-2}
$(0, 0.65)$	0.89	0.600	0.511	0.600	1.46×10^{-2}	1.46×10^{-2}	1.27×10^{-1}	1.17×10^{-1}	9.3×10^{-2} / 8.8×10^{-2}
$(0.01, 0.9)$	0.60	0.618	0.826	0.618	1.88×10^{-2}	1.88×10^{-2}	7.89×10^{-2}	7.77×10^{-2}	7.8×10^{-2} / 5.2×10^{-2}

with $e_B = 0.65$ it lowers the risk from 0.127 to 0.093, 6% above the genie (Table S7). On a 9×5 grid of the class, it wins on every channel with $b \leq 0.575$, where relapses are frequent, but loses on 23 of 45, all with $b \geq 0.7$, by up to $1.47\times$. It conditions only on the last observation, so it is not the optimal policy under ACK/NACK feedback for a second-order channel. A designer who knows the channel can evaluate both policies exactly and keep the better one. Over the grid, $R(\pi_L)/R(\pi_{\mathrm{rec}})$ ranges from 0.68 to 2.25, so the better of the two improves on π_L by up to $2.25\times$, and it lies 1.0–5.3 times above the genie. The optimal policy under ACK/NACK feedback, which requires a dynamic program over beliefs, is not computed here; its risk lies between that of the genie and that of the better of π_L and π_{rec}. On the measured links of Section S5.5, the redesign with the empirical recovery curve improves on π_L by a median factor of 1.015 and at most 1.27, but it is worse on 52 of the 171 links. These gains are in-sample, because the empirical curve is estimated from the same tests on which both policies are evaluated.

TABLE S8
DESIGNING WITH THE TRUE RECOVERY CURVE

channel	ρ	b	$R(\pi_L)$	$R(\pi_{\mathrm{rec}})$	genie
nominal	0.600	0.90	5.97×10^{-2}	5.97×10^{-2}	4.70×10^{-2}
C_1	0.440	0.95	1.30×10^{-2}	1.67×10^{-2}	6.81×10^{-3}
C_2	0.367	0.45	6.04×10^{-2}	2.68×10^{-2}	6.13×10^{-3}

S5.3 Ambiguity across links

Table S9 contains the data of Fig. 4 ($T = 60$, $\eta = 10^{-4}$) together with the worst channels, the mean burst length ℓ of the fit and the survival time D, the smallest at which the Gilbert fit certifies $R_{\mathrm{nom}} \leq 10^{-4}$ with dense attempts.

S5.4 Fading channels

Generation. Rayleigh fading with Clarke autocorrelation $J_0(2\pi f_D|t-t'|)$ between slots t and t' is generated by the inverse-DFT method of [3] on blocks of 2^{23} slots; the empirical autocorrelation matches J_0 to three decimals. A packet is lost if $|h|^2 < -\ln(1-\bar{e})$, which gives the loss rate $\bar{e}$. For composite fading, $|h|^2$ is multiplied by $10^{X_t/10}$, where X_t is a Gauss–Markov process with standard deviation 6 dB and correlation time 200 slots, and the threshold is the empirical $\bar{e}$-quantile of the received power. Four independent blocks are used per channel (3.4×10^7 slots).

Evaluation. Missions of $T = 60$ slots start every third slot; the trace-driven risk is the fraction of failing missions. Order-L fits use $(L+1)$-block frequencies of the same trace with a pseudo-count of 0.5. Table S10 gives all configurations. Its ratios are predicted/true risks for order-L Markov fits to the trace; g=2 denotes periodic attempts every second slot, and the last column gives the Gilbert and order-6 predictions for π_L ($B = 15$), which is designed on the Gilbert fit. Fig. S3 explains the errors through the run persistences of Proposition 1: the Gilbert fit extrapolates the first persistence (dotted lines), which overstates the risk where persistence first falls with depth (Rayleigh) and understates it where persistence grows with depth (shadowing and the measured links). The $M = 2$ class ($\eta = 10^{-4}$) bounds the trace-driven dense risk from above in all 15 configurations and from below in all but the three slow-Rayleigh ones with $D \geq 14$. For π_L ($B = 15$), the trace-driven risk lies below the class range at $f_D = 0.01$ and $D \geq 16$, and above R^+ at $f_D = 0.005$ and $D = 8$.

S5.5 Measured links

Data. We use the FlockLab link-quality dataset [4] (July–September 2019, TelosB, 802.15.4 channel 26, 0 dBm, 8 B payload). In each test every node sends 100 strobes, and every other node records a success/loss bit per strobe. The preprocessed files repeat each (test, sender, receiver) record several times (median $6.4\times$, always with identical content), so records are deduplicated on that key. Rows with a missing or truncated bit string and streams without any success (link down) are discarded. We keep the 171 directed links with loss rate in $[1\%, 25\%]$ and at least 500 distinct tests (in fact 567–805).

TABLE S9
AMBIGUITY ACROSS LINKS

loss	ℓ	D	R_{nom}	R^+/R_{nom}	worst ρ	R_{nom}, B=15	R^+/R_{nom}, B=15	worst (ρ, b), B=15
0.5%	1.25	6	7.1×10^{-5}	68	0.816	2.8×10^{-3}	4.1	(0.625, 0.27)
0.5%	1.5	8	8.2×10^{-5}	69	0.875	7.8×10^{-4}	10	(0.325, 0.03)
0.5%	2	12	6.1×10^{-5}	86	0.925	2.2×10^{-4}	30	(0.900, 0.28)
0.5%	2.5	15	7.4×10^{-5}	67	0.950	1.7×10^{-4}	40	(0.950, 0.05)
0.5%	3	18	7.6×10^{-5}	61	0.962	1.3×10^{-4}	39	(0.937, 0.06)
0.5%	5	28	8.9×10^{-5}	46	0.994	1.1×10^{-4}	41	(0.981, 0.10)
1%	1.25	7	2.8×10^{-5}	285	0.850	1.2×10^{-3}	14	(0.750, 0.26)
1%	1.5	9	5.3×10^{-5}	182	0.891	5.7×10^{-4}	25	(0.725, 0.01)
1%	2	13	6.0×10^{-5}	162	0.933	2.1×10^{-4}	61	(0.900, 0.02)
1%	2.5	16	8.7×10^{-5}	107	0.955	2.0×10^{-4}	53	(0.933, 0.27)
1%	3	19	9.9×10^{-5}	89	0.967	1.8×10^{-4}	59	(0.950, 0.03)
1%	5	31	8.4×10^{-5}	95	1.000	1.0×10^{-4}	81	(0.987, 0.05)
2%	1.25	7	5.6×10^{-5}	285	0.850	3.0×10^{-3}	12	(0.750, 0.25)
2%	1.5	10	3.5×10^{-5}	493	0.908	4.3×10^{-4}	62	(0.850, 0.26)
2%	2	14	5.9×10^{-5}	305	0.942	2.3×10^{-4}	95	(0.925, 0.26)
2%	2.5	18	6.0×10^{-5}	278	0.961	1.5×10^{-4}	131	(0.950, 0.26)
2%	3	21	8.4×10^{-5}	193	0.975	1.6×10^{-4}	124	(0.950, 0.01)
2%	5	34	7.9×10^{-5}	203	1.000	9.9×10^{-5}	171	(0.987, 0.02)
5%	1.25	8	2.7×10^{-5}	$1.2k$	0.875	2.9×10^{-3}	22	(0.800, 0.00)
5%	1.5	10	8.7×10^{-5}	490	0.908	1.6×10^{-3}	42	(0.850, 0.25)
5%	2	15	7.2×10^{-5}	579	0.950	3.7×10^{-4}	156	(0.925, 0.00)
5%	2.5	19	8.8×10^{-5}	451	0.967	2.7×10^{-4}	185	(0.933, 0.00)
5%	3	23	8.9×10^{-5}	426	0.979	2.0×10^{-4}	249	(0.987, 0.01)
5%	5	37	9.1×10^{-5}	440	1.000	1.3×10^{-4}	341	(0.987, 0.01)
10%	1.25	8	5.5×10^{-5}	$1.2k$	0.875	8.7×10^{-3}	14	(0.825, 0.00)
10%	1.5	11	5.7×10^{-5}	$1.3k$	0.916	1.9×10^{-3}	69	(0.875, 0.00)
10%	2	16	7.0×10^{-5}	$1.1k$	0.958	5.7×10^{-4}	170	(0.000, 0.00)
10%	2.5	21	6.1×10^{-5}	$1.2k$	0.972	2.8×10^{-4}	370	(0.983, 0.00)
10%	3	25	7.5×10^{-5}	954	0.987	2.4×10^{-4}	372	(0.975, 0.00)
10%	5	40	8.3×10^{-5}	961	1.000	1.4×10^{-4}	574	(0.994, 0.00)
20%	1.25	9	2.1×10^{-5}	$5.3k$	0.891	1.1×10^{-2}	14	(0.825, 1.00)
20%	1.5	12	3.7×10^{-5}	$3.7k$	0.933	3.6×10^{-3}	48	(0.900, 0.25)
20%	2	17	6.9×10^{-5}	$2.1k$	0.958	1.3×10^{-3}	172	(0.950, 0.00)
20%	2.5	22	7.1×10^{-5}	$2.0k$	0.978	6.6×10^{-4}	240	(0.950, 0.00)
20%	3	26	9.8×10^{-5}	$1.4k$	0.987	5.5×10^{-4}	291	(0.975, 0.00)
20%	5	42	9.8×10^{-5}	$1.6k$	1.000	2.5×10^{-4}	638	(1.000, 0.00)

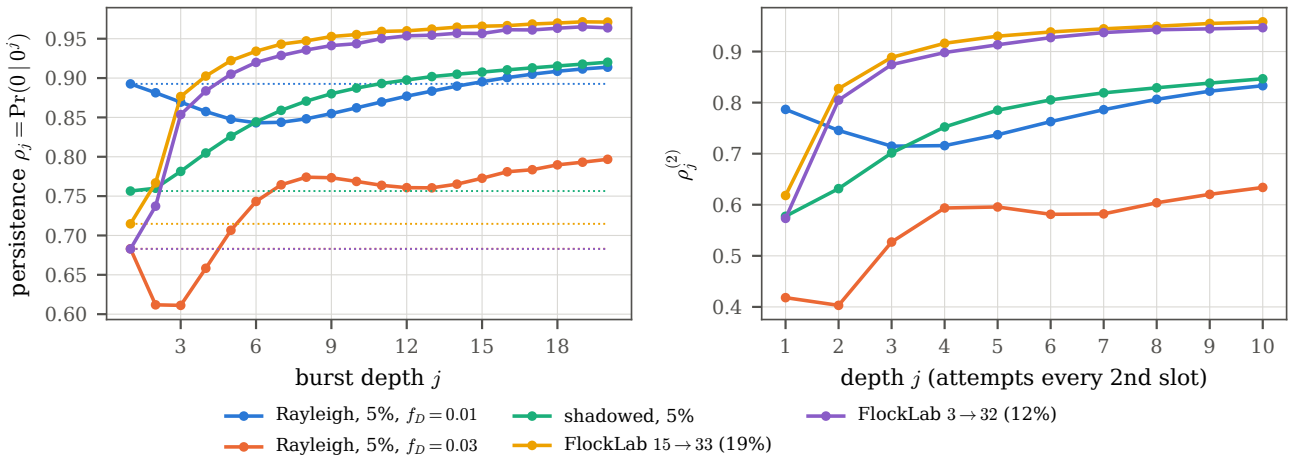


Fig. S3. Run persistences $\rho_j = \Pr(0 \mid 0^j)$ (left) and those of the 2-decimated channel (right) for fading channels and two measured links. Dotted lines: the constant persistence implied by the Gilbert fit.

TABLE S10
ALL FADING CONFIGURATIONS

channel	D	R dense	L=1	2	3	4	6	M=2 class	R, g=2	L=1	R, π_L	L=1/L=6
Rayleigh, 5%, $f_D = 0.01$	12	6.5×10^{-2}	1.3	1.3	1.2	1.2	1.2	$[4.5\times10^{-2}, 9.3\times10^{-2}]$	7.3×10^{-2}	1.3	8.4×10^{-2}	1.2 / 1.1
Rayleigh, 5%, $f_D = 0.01$	16	3.2×10^{-2}	1.6	1.5	1.4	1.2	1.1	$[3.8\times10^{-2}, 7.0\times10^{-2}]$	3.5×10^{-2}	1.6	3.7×10^{-2}	1.6 / 1.1
Rayleigh, 5%, $f_D = 0.01$	20	1.8×10^{-2}	1.7	1.5	1.3	1.1	0.93	$[2.4\times10^{-2}, 5.7\times10^{-2}]$	1.9×10^{-2}	1.7	2.0×10^{-2}	1.7 / 1
Rayleigh, 5%, $f_D = 0.03$	6	8.3×10^{-2}	1.5	1.2	1.2	1.2	1	$[3.4\times10^{-2}, 1.6\times10^{-1}]$	1.1×10^{-1}	1.5	2.1×10^{-1}	1.1 / 1
Rayleigh, 5%, $f_D = 0.03$	8	4.2×10^{-2}	1.4	0.89	0.89	1.1	1.1	$[1.8\times10^{-2}, 1.1\times10^{-1}]$	5.0×10^{-2}	1.5	7.5×10^{-2}	1.5 / 1
Rayleigh, 5%, $f_D = 0.03$	10	2.5×10^{-2}	1.1	0.54	0.54	0.74	0.99	$[5.5\times10^{-3}, 8.7\times10^{-2}]$	2.9×10^{-2}	1.2	3.7×10^{-2}	1.4 / 1.1
Rayleigh, 1%, $f_D = 0.005$	8	2.8×10^{-2}	1	1	1	1	1	$[8.7\times10^{-3}, 2.9\times10^{-2}]$	3.2×10^{-2}	0.96	3.9×10^{-2}	0.88 / 1
Rayleigh, 1%, $f_D = 0.005$	11	1.4×10^{-2}	1.4	1.3	1.2	1.2	1.1	$[8.7\times10^{-3}, 2.0\times10^{-2}]$	1.9×10^{-2}	1.2	1.8×10^{-2}	1.2 / 1.1
Rayleigh, 1%, $f_D = 0.005$	14	7.1×10^{-3}	1.7	1.5	1.3	1.2	1.1	$[8.3\times10^{-3}, 1.5\times10^{-2}]$	7.9×10^{-3}	1.6	9.0×10^{-3}	1.5 / 1.1
shadowed, 5%, $f_D = 0.03$	10	4.9×10^{-2}	1.1	1.1	1.2	1.3	1.3	$[2.5\times10^{-2}, 9.6\times10^{-2}]$	5.4×10^{-2}	1.2	6.4×10^{-2}	1.3 / 1.3
shadowed, 5%, $f_D = 0.03$	14	2.7×10^{-2}	0.6	0.62	0.78	0.94	1.2	$[5.5\times10^{-3}, 6.7\times10^{-2}]$	3.0×10^{-2}	0.66	3.3×10^{-2}	0.76 / 1.2
shadowed, 5%, $f_D = 0.03$	18	1.6×10^{-2}	0.3	0.32	0.45	0.62	0.92	$[1.1\times10^{-3}, 5.3\times10^{-2}]$	1.8×10^{-2}	0.34	1.9×10^{-2}	0.4 / 0.98
shadowed, 1%, $f_D = 0.03$	8	6.6×10^{-3}	0.44	0.71	1	1.2	1.2	$[4.2\times10^{-7}, 1.8\times10^{-2}]$	7.9×10^{-3}	0.56	1.1×10^{-2}	0.82 / 1.1
shadowed, 1%, $f_D = 0.03$	12	2.5×10^{-3}	0.086	0.21	0.46	0.73	1.1	$[5.4\times10^{-11}, 1.1\times10^{-2}]$	2.9×10^{-3}	0.11	3.5×10^{-3}	0.19 / 1.1
shadowed, 1%, $f_D = 0.03$	16	1.0×10^{-3}	0.015	0.053	0.18	0.39	0.81	$[7.0\times10^{-15}, 8.3\times10^{-3}]$	1.2×10^{-3}	0.02	1.4×10^{-3}	0.034 / 0.87

Evaluation. Each test is a short trace; missions of $T = 60$ slide over its 41 possible start positions. Fits and recovery curves are estimated from within-stream counts only. For each link, D is the smallest survival time at which the Gilbert fit certifies $R_{\text{nom}} \le 10^{-2}$ with dense attempts, and π_L ($B = 15$) is designed on that fit. Standard errors treat tests as independent units; the median relative standard error of the dense risk is 0.29, and the measured risk exceeds the Gilbert prediction by more than two standard errors on 47% of the links. It exceeds the certified 10^{-2} on 55% of the links, by more than two standard errors on 42%. Seven links have no failing mission, so their standard error is zero, and the two-standard-error rule counts all of them as significantly below the prediction. By the rule of three over tests, only four of them are, which would change the 20% of significantly lower links in Section V-B to 19%. Fig. S4 shows the whole population, and Table S11 (ℓ: mean burst length) the burstiest link of each loss band (1–4, 4–8, 8–15 and 15–25%). Its ratios are predicted/true; the last column gives the Gilbert-predicted ratio for π_L ($B = 15$) and the measured risk of the recovery-curve redesign. On all four links, the measured dense risk lies within $[R^-, R^+]$ of the $M = 2$ class ($\eta = 10^{-4}$). The error grows with loss rate and burstiness; the second-order fit underestimates on more links than the Gilbert fit (85% versus 64%), and the order-6 fit overshoots by 1.5–2.1×. The persistence profile, pooled over tests, grows with burst depth (Fig. S3), as in the shadowed channels. Part of this growth may reflect differences between tests (slow regimes) rather than memory within a test; separating the two would require fits per test or per cluster of tests.

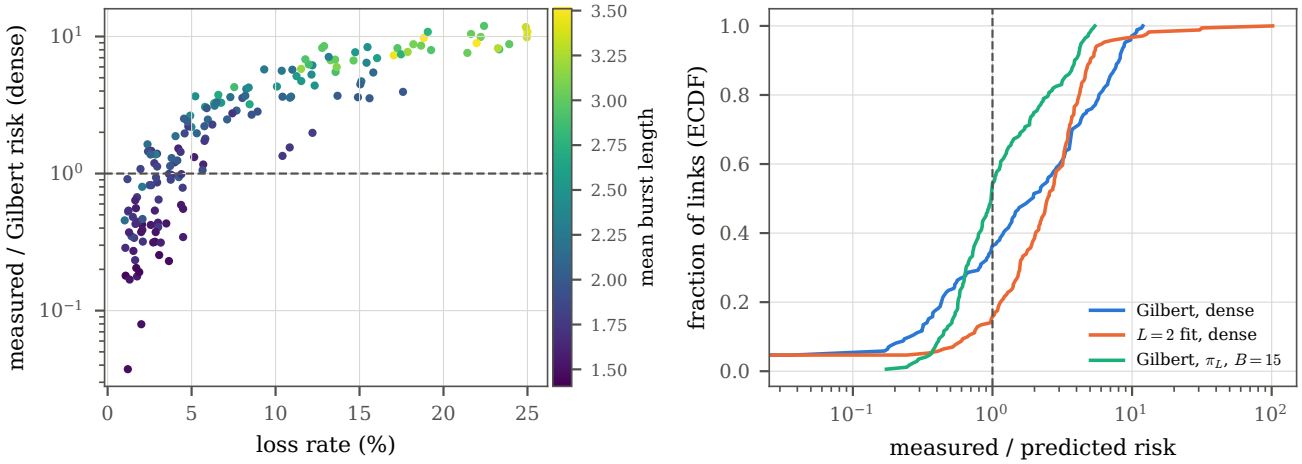


Fig. S4. Measured links (FlockLab, 171 TelosB links). Left: ratio of measured to Gilbert-predicted dense risk versus loss rate, colored by mean burst length. Right: empirical distribution of measured/predicted risk for the Gilbert fit (dense), the second-order fit (dense) and the Gilbert-predicted risk of π_L ($B = 15$).

S6. REPRODUCIBILITY

All results are computed by exact recursions over the joint chain of channel context and controller state, in floating point or exact rational arithmetic, except the trace-driven risks. Table S12 maps the claims to scripts in `code/`. The script `verify_paper_claims.py` rechecks the key numerical claims of the main paper from the stored results, and `make_figs.py` followed by `make_supplement.py` regenerates this document's tables and data figures. Designing π_L for $T = 60$, $B = 15$

TABLE S11
REPRESENTATIVE MEASURED LINKS

link	tests	loss	ℓ	D	R dense	L=1	2	3	4	6	R, π_L	L=1 / rec.
$15 \to 33$	718	18.8%	3.51	18	8.0×10^{-2}	0.1	0.26	1.1	1.3	1.5	1.1×10^{-1}	0.18 / 1.0×10^{-1}
$23 \to 22$	759	7.5%	2.86	13	3.2×10^{-2}	0.23	0.51	1.7	1.8	1.6	4.7×10^{-2}	0.37 / 4.5×10^{-2}
$27 \to 22$	801	2.9%	2.28	9	9.0×10^{-3}	0.73	1.2	3.5	3.2	2.1	1.5×10^{-2}	1.3 / 1.6×10^{-2}
$3 \to 32$	752	11.5%	3.16	15	5.0×10^{-2}	0.17	0.39	1.3	1.6	1.6	7.5×10^{-2}	0.26 / 7.1×10^{-2}

takes about 0.5 s in exact arithmetic, and evaluating one channel about 12 ms (Python, one core). The measured traces are downloaded by `get_flocklab.py` (about 50 MB).

TABLE S12
CLAIMS AND THE SCRIPTS THAT PRODUCE THEM

Claim	Script (`code/`)
Lemma 1, Eq. (12), Theorem 2, Fig. 2, Fig. S2, Tables S1 and S2	`exp_dense.py`, `exp_dense_M3.py`, `exp_phase.py`
Proposition 1, decimated persistences	`exp_decimation.py`
Eq. (14), Proposition S2, reversal criterion	`exp_analytic.py`
Eq. (15), Proposition S3	`exp_closedloop.py`
Proposition 2, exact fractions	`exp_reversal.py`
Fig. 3, Table S3, b-sensitivity, bound tightness	`exp_budget.py`
Baselines, Table S4; η, Table S5; Remark S1	`exp_baselines.py`, `exp_eta.py`, `exp_M3L1.py`
Gilbert–Elliott channels, Table S7	`ge.py`, `exp_ge.py`
Redesign, Table S8	`exp_redesign.py`
Fig. 4, Table S9	`exp_sweep.py`
Table I (fading), Table S10, Fig. S3	`exp_fading.py` (argument `shadow`)
Table I (measured), Fig. S4, Table S11	`get_flocklab.py`, `exp_flocklab.py`
Table S6	`exp_mc_check.py`

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